

★ SORTING TECHNIQUES

Sorting is a technique to rearrange the elements of a list in ascending or descending order, which can be numerical, or user-defined order.

Criteria for Analysis of Sorting Techniques —

- 1) Number of Comparisons → decides time complexity
- 2) Number of Swaps → Common process done in all sorting techniques
- 3) Adaptive → Taking less time over already sorted list
- 4) Stable → An algorithm preserving order of duplicate elements in sorted list
- 5) Extra Memory

Comparison Based Sorts	1. Bubble	} $O(n^2)$
	2. Insertion	
	3. Selection	
	4. Heap	} $O(n \log(n))$
	5. Merge	
	6. Quick	
	7. Tree	
	8. Shell — $O(n^{3/2})$	

Index- Based Sort	9. Count	} $O(n)$
	10. Bucket/Bin	
	11. Radix	